

Fundamental derivation of two Boris solvers and the Ge-Marsden theorem

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(Received 9 August 2021; accepted 20 October 2021; published 5 November 2021)

It has been known for some time that when one uses the Lorentz force law, rather than Hamilton's equation, one can derive two basic algorithms for solving trajectories in a magnetic field formally similar to the velocity-Verlet (VV) and position-Verlet (PV) symplectic integrators independent of any finite-difference approximation. Because the Lorentz force law uses the mechanical rather than the canonical momentum, the resulting magnetic field algorithms are exact energy conserving, rather than symplectic. In general, both types of algorithms can only yield the exact trajectory in the limit of vanishing small time steps. This work shows that, for a constant magnetic field, both magnetic algorithms can be further modified so that their trajectories are exactly on the gyrocircle at finite time steps. The magnetic form of the PV integrator then becomes the well-known Boris solver, while the VV form yields a second, previously unknown Boris-type algorithm, unrelated to any finite-difference scheme. Remarkably, the modification needed for the trajectory to be exact is a reparametrization of the time step, reminiscent of the Ge-Marsden theorem.

DOI: [10.1103/PhysRevE.104.055301](https://doi.org/10.1103/PhysRevE.104.055301)

I. INTRODUCTION

The Ge-Marsden [1] theorem states that if a symplectic integrator (SI) exactly conserves the energy, but no other constants of motion, then its trajectory must be exact, up to “time-reparametrization.” The requirement that there are no other constants of motion except the energy seems to restrict its applicability to one-dimensional systems. The theorem is mainly used to argue that symplectic integrators cannot be exact energy conserving. From the modern perspective of deriving SIs from Lie operators and Lie series [2–5], rather than from generating functions [6], the theorem can be viewed as a corollary of approximating the exact evolution operator as a single product:

$$e^{\Delta t \hat{H}} = e^{\Delta t (\hat{T} + \hat{V})} \approx \prod_{i=1}^N e^{a_i \Delta t \hat{T}} e^{b_i \Delta t \hat{V}} = e^{\Delta t \hat{H}'}, \quad (1)$$

where for the standard separable Hamiltonian $H = \mathbf{p}^2/2m + V(\mathbf{r})$, $\hat{T}$ and $\hat{V}$ are Lie operators,

$$\hat{T} = \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} \quad \text{and} \quad \hat{V} = \mathbf{a}(\mathbf{r}) \cdot \frac{\partial}{\partial \mathbf{v}}, \quad (2)$$

with $\mathbf{v} \equiv \mathbf{p}/m$, $\mathbf{a}(\mathbf{r}) = -\nabla V/m$, and group actions

$$\begin{aligned} e^{\Delta t \hat{T}}(\mathbf{r}, \mathbf{v}) &= (\mathbf{r} + \Delta t \mathbf{v}, \mathbf{v}), \\ e^{\Delta t \hat{V}}(\mathbf{r}, \mathbf{v}) &= (\mathbf{r}, \mathbf{v} + \Delta t \mathbf{a}). \end{aligned} \quad (3)$$

The resulting modified Hamiltonian operator

$$\hat{H}' = \hat{H} + \Delta t \hat{E}_1 + \Delta t^2 \hat{E}_2 + \Delta t^3 \hat{E}_3 + \cdots, \quad (4)$$

with various energy error terms $\hat{E}_k$, gives the trajectory of the algorithm via Hamilton's equation with respect to the Hamiltonian function [4] H' corresponding to the operator $\hat{H}'$. If

for some reasons $\hat{H}' = \hat{H}$ with all $\hat{E}_k = 0$, then the trajectory produced by H' must be the same as the exact trajectory dictated by H . From this perspective, exact energy conservation implies that exact trajectory is a general feature of symplectic integrators of the form (1), irrespective of whether there are other conserved quantities. This result only requires the existence of the modified Hamiltonian and its associated dynamics.

Moreover, the original proof of Ge and Marsden [1] allows for “time reparametrization” of the form [6] $H' = F(\Delta t)H$, which usually does not occur in practice. This is because H' as given by (4) must have $F(\Delta t) = 1$ due to the consistency requirements:

$$\sum_{i=1}^N a_i = 1 \quad \text{and} \quad \sum_{i=1}^N b_i = 1. \quad (5)$$

This can be easily verified by applying the following second-order velocity-Verlet SI:

$$\mathbf{v}_1 = \mathbf{v}_0 + \frac{1}{2} \Delta t \mathbf{a}(\mathbf{r}_0), \quad (6)$$

$$\mathbf{r}_1 = \mathbf{r}_0 + \Delta t \mathbf{v}_1, \quad (7)$$

$$\mathbf{v}_2 = \mathbf{v}_1 + \frac{1}{2} \Delta t \mathbf{a}(\mathbf{r}_1), \quad (8)$$

corresponding to $N = 2$, $a_1 = 0$, $b_1 = 1/2$, $a_2 = 1$, and $b_2 = 1/2$ in (1), and the position-Verlet SI:

$$\begin{aligned} \mathbf{r}_1 &= \mathbf{r}_0 + \frac{1}{2} \Delta t \mathbf{v}_0, \\ \mathbf{v}_1 &= \mathbf{v}_0 + \Delta t \mathbf{a}(\mathbf{r}_1), \end{aligned} \quad (9)$$

$$\mathbf{r}_2 = \mathbf{r}_1 + \frac{1}{2} \Delta t \mathbf{v}_1, \quad (10)$$

corresponding to $N = 2$, $a_1 = 1/2$, $b_1 = 1$, $a_2 = 1/2$, and $b_2 = 0$ in (1), to the case of a *constant* force with $\mathbf{a}(\mathbf{r}) = \mathbf{a}$. Both algorithms then give the final positions (7) and (10)

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as $\mathbf{r}_{1,2}(\Delta t) = \mathbf{r}_0 + \Delta t \mathbf{v}_0 + \frac{1}{2} \Delta t^2 \mathbf{a}$, which is the exact trajectory for all Δt with no time reparametrization. The author is unaware of any example of the Ge-Marsden theorem, where the integrator is exact energy conserving and the trajectory is exactly correct, but only in terms of time $\tau = \tau(t)$, which is a reparametrized function of the actual time t . It is, therefore, a complete surprise that a highly nontrivial time-reparametrization is needed to obtain exact trajectories in a *constant* magnetic field. A resulting algorithm from such a time reparametrization is none other than the well-known Boris solver of plasma physics.

The Boris solver [7–9] has been widely regarded as the gold standard for solving charged particle trajectories in plasma [9] and astrophysics [10]. But why is it so good [11]? Recent works focusing on volume-preserving methods [11,12] do not explain why the Boris solver is unique among all volume-preserving algorithms. The real reason why the Boris solver is so good, according to Buneman [7], who proposed an earlier version of the solver, is that “it leads to points on exact cycloids whenever $\mathbf{E}$ and $\mathbf{B}$ are constant.” That is, for a constant magnetic field, positions generated by the Buneman solver [7] are exactly on the gyrocircle, the cyclotron orbit. However, when Buneman’s original $\mathbf{E} \times \mathbf{B}$ formulation was abandoned [9] in favor of Boris’s $\mathbf{E}$, $\mathbf{B}$ splitting scheme, this point seems to have been forgotten, especially when only the first-order leap-frog version of the Boris solver is used [9,13–15]. Otherwise, this on-orbit property need not be pointed out again by others [16] when using the second-order solver [15].

The same operator method used to derive the symplectic integrator above has been used by the author to derive *exact energy conserving* (EEC) integrators for solving charged particle trajectories in a general magnetic field [17]. Since these algorithms use the Lorentz force law directly in terms of the *mechanical* momentum rather than the *canonical* momentum, these algorithms are EEC but not symplectic. (They are referred to as *Poisson integrators* in Ref. [18].) Because of this, their properties are not directly related to the Ge-Marsden theorem: they conserved the energy exactly, yet their trajectories are not exact. For a constant magnetic field, their orbits are not on the gyrocircle. According to Buneman [7], that can only be achieved with additional “cycloid fitting.” Remarkably, this work shows that this “cycloid fitting” can be done exactly by a time-step reparametrization, a possibility suggested by the Ge-Marsden theorem. One of the resulting algorithms is then the Boris solver.

This work presents a completely original derivation of Boris solvers independent of any finite-difference schemes, including a previously unknown, second Boris solver. To see how these two Boris solvers are related to symplectic integrators, this work reviews in the next section the magnetic velocity-Verlet and position-Verlet integrators previously derived in Ref. [17]. In Sec. III, the two Boris solvers are derived by “cycloid fitting.” Simple higher-order algorithms are then constructed in Sec. IV. Conclusions are stated in Sec. V.

II. THE TWO FUNDAMENTAL MAGNETIC ALGORITHMS

For a charged particle q in a magnetic field $\mathbf{B}(\mathbf{r}) = B(\mathbf{r})\hat{\mathbf{B}}(\mathbf{r})$, with magnitude $B(\mathbf{r})$ and unit vector $\hat{\mathbf{B}}(\mathbf{r})$, its ac-

celeration is given by the Lorentz force law:

$$\frac{d\mathbf{v}}{dt} = \omega(\mathbf{r})\hat{\mathbf{B}}(\mathbf{r}) \times \mathbf{v}, \quad (11)$$

where $\omega(\mathbf{r}) = (-q)B(\mathbf{r})/m$ is the local cyclotron angular frequency. (Note that ω is correctly positive for $q < 0$ because if $\hat{\mathbf{B}}(\mathbf{r})$ is out of the page, the cyclotron motion is counterclockwise.) The operator $\hat{V}$ in (2) is then replaced by

$$\hat{V}_B = \omega(\hat{\mathbf{B}} \times \mathbf{v}) \cdot \frac{\partial}{\partial \mathbf{v}}, \quad (12)$$

whose effect on $\mathbf{v}$ is just

$$\begin{aligned} \hat{V}_B \mathbf{v} &= (\omega \hat{\mathbf{B}} \times) \mathbf{v}, \\ \hat{V}_B^n \mathbf{v} &= (\omega \hat{\mathbf{B}} \times)^n \mathbf{v}, \end{aligned} \quad (13)$$

and therefore, the velocity update is

$$\mathbf{v}_B(\mathbf{r}, \mathbf{v}, \Delta t) = e^{\Delta t \hat{V}_B} \mathbf{v} = e^{\theta(\hat{\mathbf{B}} \times)} \mathbf{v} \quad (14)$$

$$\begin{aligned} &= \mathbf{v} + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots \right) \hat{\mathbf{B}} \times \mathbf{v} \\ &\quad + \left(\frac{\theta^2}{2} - \frac{\theta^4}{4!} + \cdots \right) \hat{\mathbf{B}} \times (\hat{\mathbf{B}} \times \mathbf{v}) \\ &= \mathbf{v} + \sin \theta \hat{\mathbf{B}} \times \mathbf{v} + (1 - \cos \theta) \hat{\mathbf{B}} \times (\hat{\mathbf{B}} \times \mathbf{v}), \end{aligned} \quad (15)$$

where $\theta = \omega(\mathbf{r})\Delta t$ and where one has repeatedly used the identity

$$\hat{\mathbf{B}} \times [\hat{\mathbf{B}} \times (\hat{\mathbf{B}} \times \mathbf{v})] = -\hat{\mathbf{B}} \times \mathbf{v} \quad (17)$$

to reduce the exponential series to (15) and resummed as (16), which is the exact solution to (11) holding $\mathbf{r}$ fixed. By decomposing $\mathbf{v}$ into components parallel and perpendicular to $\hat{\mathbf{B}}(\mathbf{r})$,

$$\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp}, \quad (18)$$

so that $\hat{\mathbf{B}} \times (\hat{\mathbf{B}} \times \mathbf{v}_{\perp}) = -\mathbf{v}_{\perp}$, (16) becomes

$$\mathbf{v}_B(\mathbf{r}, \mathbf{v}, \Delta t) = \mathbf{v}_{\parallel} + \cos \theta \mathbf{v}_{\perp} + \sin \theta (\hat{\mathbf{B}} \times \mathbf{v}_{\perp}). \quad (19)$$

This shows that the magnetic field only rotates the perpendicular velocity component without changing the velocity’s magnitude and hence the kinetic energy.

By replacing the velocity updates in (6), (8), and (9) by (16) or (19), one obtains the corresponding magnetic integrator, named here as M2a, given below:

$$\mathbf{v}_1 = \mathbf{v}_B(\mathbf{r}_0, \mathbf{v}_0, \Delta t/2), \quad (20)$$

$$\mathbf{r}_1 = \mathbf{r}_0 + \Delta t \mathbf{v}_1, \quad (21)$$

$$\mathbf{v}_2 = \mathbf{v}_B(\mathbf{r}_1, \mathbf{v}_1, \Delta t/2), \quad (22)$$

and M2b as given by

$$\begin{aligned} \mathbf{r}_1 &= \mathbf{r}_0 + \frac{1}{2} \Delta t \mathbf{v}_0, \\ \mathbf{v}_1 &= \mathbf{v}_B(\mathbf{r}_1, \mathbf{v}_0, \Delta t), \end{aligned} \quad (23)$$

$$\mathbf{r}_2 = \mathbf{r}_1 + \frac{1}{2} \Delta t \mathbf{v}_1. \quad (24)$$

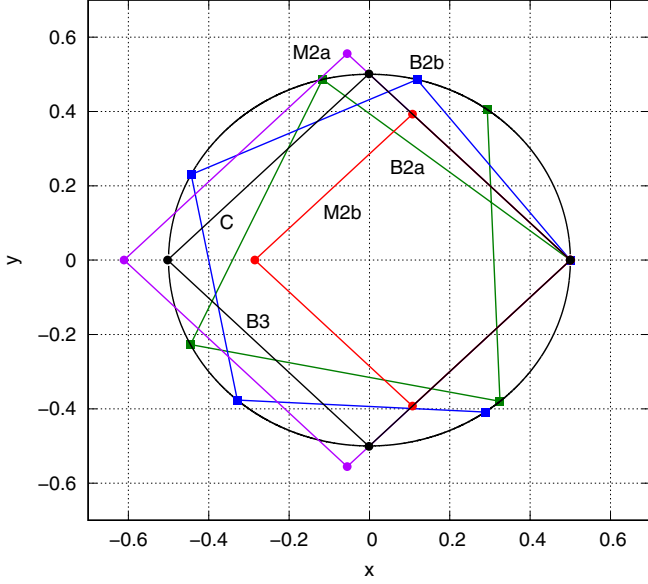


FIG. 1. All orbits start on the rightmost point on the gyrocircle. M2a and M2b have three points outside and inside of the circle, respectively. B2b and B2a have points on the circle but with angles behind and ahead of the phase correct angle of M2a and M2b. C and B3 are higher-order results having very small error off the circle and very small phase error, respectively. See text for details.

These two algorithms have been previously derived in Ref. [17] and were named then simply as 2a and 2b. Each is completely explicit, sequential, and exact energy conserving, but *not* symplectic. This is because in order to be symplectic, one must update the canonical momentum in Hamilton's equations of motion. Since the Lorentz force law (11) uses only the mechanical momentum, any direct use of the Lorentz force law is noncanonical and therefore nonsymplectic.

To see how these magnetic integrators work, consider the case of an electron in a constant magnetic field with $\omega = 2$, $\hat{\mathbf{B}} = \hat{\mathbf{z}}$, $\mathbf{r} = (x, y)$, and $\mathbf{v} = \mathbf{v}_\perp = (v_x, v_y)$, with initial velocity $\mathbf{v}_0 = (0, v_0)$ and $\mathbf{r}_0 = (r_g, 0)$, where $v_0 = 1$ and where $r_g = v_0/\omega = 1/2$ is the gyroradius. Let us take a large $\Delta t = \pi/4$, so that $\theta = \omega\Delta t = \pi/2$. Therefore, both $\mathbf{r}$ and $\mathbf{v}$ rotate through 90° 4 times to complete one orbit. The M2a and M2b trajectories are the square orbits shown in Fig. 1. For M2a, the position update (21) is Δt times the initial velocity rotated 45° , so the length of the displacement is $v_0\Delta t$. For M2b, the position update (24) is Δt times half the diagonal of a square with sides v_0 , which is $v_0\Delta t/\sqrt{2}$. Therefore, the displacement of M2a is $\sqrt{2}$ times that of M2b, as shown. Their trajectories fall outside and inside of the gyrocircle, respectively.

III. CYCLOID FITTING

For M2b, the position updating step (24) is

$$\begin{aligned} \mathbf{r}_2 &= \mathbf{r}_0 + \frac{1}{2}\Delta t(\mathbf{v}_0 + \mathbf{v}_1) \\ &= \mathbf{r}_0 + \frac{1}{2}\Delta t(\mathbf{v}_0 + \cos\theta\mathbf{v}_0 + \sin\theta\hat{\mathbf{B}} \times \mathbf{v}_0). \end{aligned} \quad (25)$$

Defining $\mathbf{r}_g = \mathbf{v}_0/\omega = (0, r_g)$ and noting that $\mathbf{r}_0 = (r_g, 0) = -\hat{\mathbf{B}} \times \mathbf{r}_g$, one can rewrite the above as

$$\mathbf{r}_2 = \frac{\theta}{2}(1 + \cos\theta)\mathbf{r}_g - \left(1 - \frac{\theta}{2}\sin\theta\right)\hat{\mathbf{B}} \times \mathbf{r}_g, \quad (26)$$

and therefore, the trajectory is always inside of the gyrocircle:

$$|\mathbf{r}_2|^2 = r_g^2 \left\{ 1 - 2\theta \cos^2\left(\frac{\theta}{2}\right) \left[\tan\left(\frac{\theta}{2}\right) - \frac{\theta}{2} \right] \right\} \quad (27)$$

$$= r_g^2 \left[1 - \frac{1}{12}\theta^4 + \frac{1}{80}\theta^6 + \dots \right] \text{ as } \theta \rightarrow 0. \quad (28)$$

However, if one were to *decouple* the actual rotation angle (arguments of trig functions) from θ to an effective angle θ_B , then requiring

$$\tan\left(\frac{\theta_B}{2}\right) = \frac{\theta}{2} \quad (29)$$

in (27) would force the trajectory to be exactly on the gyrocircle. This corresponds to a highly nontrivial reparametrization of Δt in the rotation angle to a Boris time step Δt_B via

$$\Delta t_B = \frac{2}{\omega} \tan^{-1}(\omega\Delta t/2). \quad (30)$$

If the updating rotation step (19) is replaced by

$$\mathbf{v}_B(\mathbf{r}, \mathbf{v}, \Delta t) = \mathbf{v}_\parallel + \cos\theta_B\mathbf{v}_\perp + \sin\theta_B(\hat{\mathbf{B}} \times \mathbf{v}_\perp), \quad (31)$$

then M2b reproduces the Boris solver denoted here as B2b. From (29), $\cos\theta_B$ and $\sin\theta_B$ can be easily reconstructed as

$$\sin\theta_B = \frac{2\tan(\theta_B/2)}{1 + \tan^2(\theta_B/2)} = \frac{\theta}{1 + \theta^2/4}, \quad (32)$$

$$\cos\theta_B = \frac{1 - \tan^2(\theta_B/2)}{1 + \tan^2(\theta_B/2)} = \frac{1 - \theta^2/4}{1 + \theta^2/4}. \quad (33)$$

This makes it clear that the position vector (26),

$$\mathbf{r}_2 = \frac{\theta}{1 + \frac{1}{4}\theta^2}\mathbf{r}_g - \frac{1 - \frac{1}{4}\theta^2}{1 + \frac{1}{4}\theta^2}\hat{\mathbf{B}} \times \mathbf{r}_g \quad (34)$$

$$\begin{aligned} &= \sin(\theta_B)\mathbf{r}_g - \cos(\theta_B)\hat{\mathbf{B}} \times \mathbf{r}_g \\ &= \cos(\theta_B - \pi/2)\mathbf{r}_g + \sin(\theta_B - \pi/2)\hat{\mathbf{B}} \times \mathbf{r}_g, \end{aligned} \quad (35)$$

is not only precisely on the gyrocircle, but it lags 90° behind, and is therefore perpendicular to the Boris-angle rotated velocity vector (31).

The resulting trajectory in Fig. 1 is labeled as B2b. Notice that B2b's angle of rotation lags behind that of M2a and M2b. This follows from $\theta = \pi/2 = 1.5708$,

$$\theta_B = 2\tan^{-1}(\theta/2) = 1.3316 \quad \text{and} \quad \theta_B - \theta = -0.2392. \quad (36)$$

More generally at small θ , the Boris angle θ_B has a second-order phase error,

$$\theta_B = \theta \left[1 - \frac{1}{12}\theta^2 + \frac{1}{80}\theta^4 + \dots \right], \quad (37)$$

showing B2b's rotation angle always lags, but remarkably with the same two leading coefficients as in the gyroradius (28). For extensive comparisons of B2b with M2a and M2b

on *nonuniform* magnetic fields, see the work by Knapp *et al.* [18].

Since θ_B can be defined via (29) for all Δt , B2b's trajectory will stay on the gyrocircle for all Δt , no matter how large. In the limit of $\theta \rightarrow \infty$ and $\theta_B \rightarrow \pi$, B2b's trajectory will just bounce back and forth nearly as straight lines across the diameter of the gyrocircle. Nevertheless, this means that B2b's trajectory remains bounded even as $\Delta t \rightarrow \infty$. It is unusual that an explicit algorithm can be unconditionally stable.

Conventionally, the stability of the original Boris solver was thought to be connected to the Cayley approximation for the exponential operator in (14):

$$e^{\theta(\hat{\mathbf{B}} \times)} \approx \left[1 - \frac{1}{2}\theta(\hat{\mathbf{B}} \times)\right]^{-1} \left[1 + \frac{1}{2}\theta(\hat{\mathbf{B}} \times)\right] \equiv \mathbf{C}, \quad (38)$$

found in the implicit midpoint approximation used in Boris's original derivation [8]. By expanding the inverse operator as $1/(1-x) = 1+x+x^2+\dots$ and recollecting terms as in (15), one finds the same Boris rotation:

$$\begin{aligned} \mathbf{C}\mathbf{v} &= \left[1 + \frac{\frac{1}{2}\theta(\hat{\mathbf{B}} \times) + \frac{1}{4}\theta^2(\hat{\mathbf{B}} \times)^2}{1 + \frac{1}{4}\theta^2}\right] \left[1 + \frac{1}{2}\theta(\hat{\mathbf{B}} \times)\right] \mathbf{v}, \\ &= \left[1 + \frac{\theta(\hat{\mathbf{B}} \times) + \frac{1}{2}\theta^2(\hat{\mathbf{B}} \times)^2}{1 + \frac{1}{4}\theta^2}\right] \mathbf{v}, \\ &= \mathbf{v}_{\parallel} + \frac{1 - \frac{1}{4}\theta^2}{1 + \frac{1}{4}\theta^2} \mathbf{v}_{\perp} + \frac{\theta}{1 + \frac{1}{4}\theta^2} (\hat{\mathbf{B}} \times \mathbf{v}_{\perp}). \end{aligned} \quad (39)$$

However, by itself, this Boris rotation (39), no more than the exact rotation (19), can confer greater stability. That is, if this Boris rotation (39) is just another *mathematical* means of rotating the velocity vector (but without evaluating trigonometric functions), then there is no *a priori* reason to expect that it would work better than the exact rotation (19). However, this derivation showed that the Boris angle condition (29) forces the trajectory to be on the gyrocircle. It is by enforcing this *physical* property that the Boris rotation can achieve greater, and in this case unconditional, stability. In principle, the angle condition (29) is totally sufficient to define the Boris rotation (31) via (32) and (33). It is entirely irrelevant whether this angle condition can be realized by any specific approximation of the exponential operator such as the Cayley transform (38).

For M2a, the position updating step (21) is

$$\mathbf{r}_1 = \mathbf{r}_0 + \Delta t [\cos(\theta/2) \mathbf{v}_0 + \sin(\theta/2) \hat{\mathbf{B}} \times \mathbf{v}_0] \quad (40)$$

$$= \theta \cos(\theta/2) \mathbf{r}_g + [\theta \sin(\theta/2) - 1] \hat{\mathbf{B}} \times \mathbf{r}_g, \quad (41)$$

$$\begin{aligned} |\mathbf{r}_1|^2 &= r_g^2 \left\{ 1 + 2\theta \left[\frac{\theta}{2} - \sin\left(\frac{\theta}{2}\right) \right] \right\} \\ &= r_g^2 \left[1 + \frac{1}{24}\theta^4 - \frac{1}{1920}\theta^6 + \dots \right] \text{ as } \theta \rightarrow 0. \end{aligned} \quad (42)$$

Therefore, if one now defines a second Boris angle θ'_B by

$$\sin\left(\frac{\theta'_B}{2}\right) = \frac{\theta}{2} \quad (43)$$

and sets

$$\cos\left(\frac{\theta'_B}{2}\right) = \sqrt{1 - \left(\frac{\theta}{2}\right)^2} \quad (44)$$

in (20) and (22), then one obtains a second, previously unknown, Boris solver whose trajectory is also exactly on the gyrocircle. The position vector (41) is then

$$\mathbf{r}_1 = \theta \sqrt{1 - \left(\frac{\theta}{2}\right)^2} \mathbf{r}_g - (1 - \theta^2/2) \hat{\mathbf{B}} \times \mathbf{r}_g \quad (45)$$

$$\begin{aligned} &= \sin(\theta'_B) \mathbf{r}_g - \cos(\theta'_B) \hat{\mathbf{B}} \times \mathbf{r}_g \\ &= \cos(\theta'_B - \pi/2) \mathbf{r}_g + \sin(\theta'_B - \pi/2) \hat{\mathbf{B}} \times \mathbf{r}_g, \end{aligned} \quad (46)$$

which is again 90° behind the θ'_B rotated velocity.

This second Boris solver's orbit is labeled as B2a in Fig. 1. Here, θ'_B is ahead of M2a and M2b by an amount nearly equal to that of (36):

$$\theta'_B = 2 \sin^{-1}(\theta/2) = 1.8067 \quad \text{and} \quad \theta'_B - \theta = 0.2359. \quad (47)$$

At small θ , θ'_B remains ahead, with the second-order error

$$\theta'_B = \theta \left[1 + \frac{1}{24}\theta^2 + \frac{3}{640}\theta^4 + \dots \right] \quad (48)$$

having the same leading coefficient as in (42), but not the exact opposite of (37).

From (43), this θ'_B can only be defined for $|\theta/2| \leq 1$. Therefore, this second Boris solver is limited to $|\Delta t| < |2/\omega| = T/\pi$, approximately a third of the gyroperiod. This same range of stability is obtained when the symplectic VV integrator is applied to the harmonic oscillator [5].

This second Boris solver also underscores the fact it is the second Boris angle condition (43) that is of paramount importance. The use of the Cayley transform (38) here would not have improved the M2a integrator in any way. Also, it is not clear whether this second Boris condition (43) can be realized by a Cayley-like operator approximation of the exponential, as in (38).

Finally, note that for motion perpendicular to $\hat{\mathbf{B}}$, since

$$\frac{d}{dt}(\mathbf{r}^2) = 2\mathbf{r} \cdot \mathbf{v} \quad \text{and} \quad \frac{d(\mathbf{r} \times \mathbf{v})}{dt} = \omega(\mathbf{r} \cdot \mathbf{v}) \hat{\mathbf{B}}, \quad (49)$$

the constancy of r^2 , the orthogonality of $\mathbf{v}$ to $\mathbf{r}$, and the conservation of angular momentum are one and the same condition.

IV. HIGHER-ORDER ALGORITHMS

Given basic second-order algorithms M2a, M2b and B2b, B2a, well-known methods [12,17,18] can be used to generate higher-order schemes. However, some schemes are so simple that they can be constructed by inspection. For example, in view of (28) and (42), the combination algorithm

$$\mathbf{C}(\Delta t) = \frac{2}{3} \mathbf{M}2a(\Delta t) + \frac{1}{3} \mathbf{M}2b(\Delta t) \quad (50)$$

would have sixth-order r_g^2 error at small Δt :

$$|\mathbf{r}|^2 = r_g^2 \left[1 + \frac{11}{2880}\theta^6 + \dots \right]. \quad (51)$$

Surprisingly, it is already very good even at $\Delta t = \pi/4$. This is shown as the square orbit C in Fig. 1. As pointed out previously, the length of the M2a step is $v_0 \Delta t = \pi/4$, and the length of the M2b step is $(\pi/4)/\sqrt{2}$. The length of the

combination C step is, therefore,

$$\frac{2\pi}{3} + \frac{1}{3} \frac{\pi}{4} \frac{1}{\sqrt{2}} = \frac{\pi}{12} (2\sqrt{2} + 1) \frac{1}{\sqrt{2}} \approx 1.002 \frac{1}{\sqrt{2}}. \quad (52)$$

Since the length require to be on the gyrocircle is $1/\sqrt{2}$, algorithm C's slight over-shoot is not visually visible. Since C has no phase error, it is very close to, but not exactly on the gyrocircle.

For B2b and B2a, since their trajectories are already on the gyrocircle, one can only improve their phase error. To preserve their positions on the gyrocircle, one must apply B2b and B2a sequentially. In view of (37) and (48), a time-symmetric sequence is

$$B3(\Delta t) = B2a\left(\frac{\Delta t}{3}\right) B2b\left(\frac{\Delta t}{3}\right) B2a\left(\frac{\Delta t}{3}\right), \quad (53)$$

with the Boris angle

$$\theta_{B3} = 4 \sin^{-1}(\theta/6) + 2 \tan^{-1}(\theta/6) = \theta \left[1 + \frac{7}{77760} \theta^4 \dots \right]. \quad (54)$$

For the case considered, $\theta = \pi/2$,

$$\theta_{B3} = 1.0005 \left(\frac{\pi}{2} \right),$$

which is, surprisingly, already in agreement with the small angle estimate (54). The resulting orbit, labeled B3, overlaps with orbit C in Fig. 1.

V. CONCLUSIONS AND FUTURE DIRECTIONS

In this work, two Boris solvers are derived from two fundamental magnetic field integrators, M2b and M2a. This derivation demonstrated the underlying unity among all second-order explicit algorithms, from symplectic, magnetic

to Boris integrators. There is no need to tinker with various finite-difference schemes or concoct *ad hoc* adjustments. There are only two fundamental algorithms in all cases, from which higher-order schemes can be systematically constructed.

As shown in the Introduction, a constant force, including that from a constant electric field, can be exactly integrated by a second-order algorithm. Therefore, for a constant electric field and a constant magnetic field, cycloid fitting, as shown by the Buneman solver [7], remains just fitting the gyrocircle. The added electric field can be easily accounted for by splitting [8,18].

Boris originally suggested [8,16] the $\tan(\alpha)/\alpha$ modification, which is to replace $\theta/2$ by $\tan(\theta/2)$, so that his solver's phase would be correct by retracing (32) and (33) in the reverse order. However, this work showed that such a modification would simply revert B2b back to M2b, losing large Δt stability and going off the gyrocircle.

From (3), the characteristics of symplectic integrators are position and velocity translations. From (19), the characteristics of magnetic integrators are position translations and velocity rotations. They are, therefore, fundamentally different. For symplectic integrators, the Ge-Marsden theorem correctly singles out the algorithm's energy, the modified Hamiltonian, as the controlling factor in determining the trajectory's accuracy. For purely magnetic integrators, because they are not Hamiltonian based, the controlling factor is not known. It is not clear what plays the role equivalent to the modified Hamiltonian. This work showed that exact energy conserving magnetic integrators must have reparametrized time steps, or phase errors, in order to produce exact trajectories. Yet, the derivation is just retrofitting, one has no theory of how to exploit the phase freedom to achieve greater trajectory accuracy. All these suggest that an equivalent Ge-Marsden theorem for magnetic integrators would be most helpful in guiding the future development of more efficient magnetic algorithms.

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